

1.Program

```
import re

resume = """

Name: Rahul Sharma

Email: rahul@gmail.com

Mobile: 9876543210

Skills: Python, SQL, Machine Learning, NLP

Experience: 3 years

"""

name = re.search(r"Name:\s*(.*)", resume)

email = re.search(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", resume)

mobile = re.search(r"\b[6-9]\d{9}\b", resume)

skills = ["Python", "Java", "SQL", "Machine Learning", "NLP"]

found_skills = []

for skill in skills:

    if re.search(skill, resume, re.IGNORECASE):

        found_skills.append(skill)

experience = re.search(r"(\d+)\s+years", resume)

years = int(experience.group(1))

print("----- Candidate Summary -----")

print("Name :", name.group(1))

print("Email :", email.group())

print("Mobile :", mobile.group())

print("Skills :", " , ".join(found_skills))

print("Experience :", years, "Years")

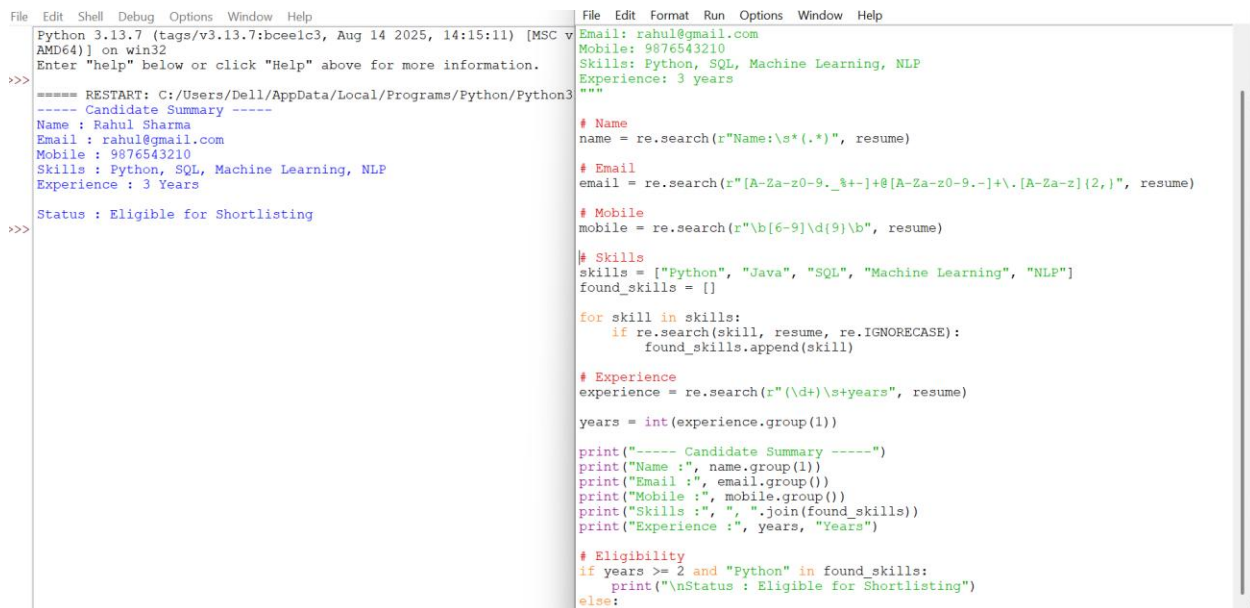
if years >= 2 and "Python" in found_skills:

    print("\nStatus : Eligible for Shortlisting")
```

else:

```
print("\nStatus : Not Eligible")
```

Output:



The screenshot shows a Python IDE with two windows. The left window displays the execution output of a script, and the right window shows the source code of the script.

Left Window (Output):

```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python3
----- Candidate Summary -----
Name : Rahul Sharma
Email : rahul@gmail.com
Mobile : 9876543210
Skills : Python, SQL, Machine Learning, NLP
Experience : 3 Years

Status : Eligible for Shortlisting
>>>
```

Right Window (Code):

```
File Edit Format Run Options Window Help

Email: rahul@gmail.com
Mobile: 9876543210
Skills: Python, SQL, Machine Learning, NLP
Experience: 3 years
"""

# Name
name = re.search(r"Name:\s*(.*)", resume)

# Email
email = re.search(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", resume)

# Mobile
mobile = re.search(r"\b[6-9]\d{9}\b", resume)

# Skills
skills = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
found_skills = []

for skill in skills:
    if re.search(skill, resume, re.IGNORECASE):
        found_skills.append(skill)

# Experience
experience = re.search(r"(\d+)\s+years", resume)
years = int(experience.group(1))

print("----- Candidate Summary -----")
print("Name :", name.group(1))
print("Email :", email.group())
print("Mobile :", mobile.group())
print("Skills :", ", ".join(found_skills))
print("Experience :", years, "Years")

# Eligibility
if years >= 2 and "Python" in found_skills:
    print("\nStatus : Eligible for Shortlisting")
else:
```

2.Program

```
import re
```

```
products = [
```

```
    "Apple iPhone",
```

```
    "Apple Watch",
```

```
    "Samsung Galaxy",
```

```
    "Galaxy Buds",
```

```
    "Redmi Note",
```

```
    "Dell Laptop",
```

```
    "HP Laptop",
```

```
"Wireless Mouse",
"Gaming Keyboard"
]
keyword = input("Enter Search Keyword: ")
print("\nExact Match")
exact = [p for p in products if re.fullmatch(keyword, p, re.IGNORECASE)]
print(exact)
print("\nPrefix Match")
prefix = [p for p in products if re.match(keyword, p, re.IGNORECASE)]
print(prefix)
print("\nSuffix Match")
suffix = [p for p in products if re.search(keyword + r"$", p, re.IGNORECASE)]
print(suffix)
print("\nPartial Match")
partial = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
print(partial)
print("\nCase-Insensitive Match")
case = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
print(case)
print("\n----- Report -----")
print("Exact Matches :", len(exact))
print("Prefix Matches :", len(prefix))
print("Suffix Matches :", len(suffix))
print("Partial Matches :", len(partial))
print("Case-Insensitive Matches :", len(case))
```

Output:

```
File Edit Shell Debug Options Window Help Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python3
Enter Search Keyword: Laptop

Exact Match
[]

Prefix Match
[]

Suffix Match
['Dell Laptop', 'HP Laptop']

Partial Match
['Dell Laptop', 'HP Laptop']

Case-Insensitive Match
['Dell Laptop', 'HP Laptop']

----- Report -----
Exact Matches : 0
Prefix Matches : 0
Suffix Matches : 2
Partial Matches : 2
Case-Insensitive Matches : 2
>>>
```

```
File Edit Format Run Options Window Help
import re

products = [
    "Apple iPhone",
    "Apple Watch",
    "Samsung Galaxy",
    "Galaxy Buds",
    "Redmi Note",
    "Dell Laptop",
    "HP Laptop",
    "Wireless Mouse",
    "Gaming Keyboard"
]

# Search Keyword
keyword = input("Enter Search Keyword: ")

print("\nExact Match")
exact = [p for p in products if re.fullmatch(keyword, p, re.IGNORECASE)]
print(exact)

print("\nPrefix Match")
prefix = [p for p in products if re.match(keyword, p, re.IGNORECASE)]
print(prefix)

print("\nSuffix Match")
suffix = [p for p in products if re.search(keyword + r"$", p, re.IGNORECASE)]
print(suffix)

print("\nPartial Match")
partial = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
print(partial)

print("\nCase-Insensitive Match")
case = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
print(case)

print("\n----- Report -----")
print("Exact Matches :", len(exact))
print("Prefix Matches :", len(prefix))
```

3.Program

```
import re

reg_no = input("Enter Register Number: ")

email = input("Enter College Email: ")

course = input("Enter Course Code: ")

semester = input("Enter Semester: ")

mobile = input("Enter Mobile Number: ")

valid = True

if re.fullmatch(r"\d{2}[A-Z]{2}\d{3}", reg_no):

    print("Register Number : Valid")

else:

    print("Register Number : Invalid")

valid = False
```

```
if re.fullmatch(r"[a-zA-Z0-9._%+-]+@college\.edu", email):
```

```
    print("Email : Valid")
```

```
else:
```

```
    print("Email : Invalid")
```

```
    valid = False
```

```
if re.fullmatch(r"[A-Z]{2}\d{3}", course):
```

```
    print("Course Code : Valid")
```

```
else:
```

```
    print("Course Code : Invalid")
```

```
    valid = False
```

```
if re.fullmatch(r"[1-8]", semester):
```

```
    print("Semester : Valid")
```

```
else:
```

```
    print("Semester : Invalid")
```

```
    valid = False
```

```
if re.fullmatch(r"[6-9]\d{9}", mobile):
```

```
    print("Mobile Number : Valid")
```

```
else:
```

```
    print("Mobile Number : Invalid")
```

```
    valid = False
```

```
print("\n----- Registration Report -----")
```

```
if valid:
```

```
    print("Registration Successful")
```

```
else:
```

```
    print("Registration Failed")
```

Output:

```
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.194
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/n
Enter Register Number: 22CS101
Enter College Email: student@college.edu
Enter Course Code: CS301
Enter Semester: 5
Enter Mobile Number: 9876543210
Register Number : Valid
Email : Valid
Course Code : Invalid
Semester : Valid
Mobile Number : Valid

----- Registration Report -----
Registration Failed
>>

File Edit Format Run Options Window Help
import re

reg_no = input("Enter Register Number: ")
email = input("Enter College Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

valid = True

# Register Number (Example: 22CS101)
if re.fullmatch(r"\d{2}[A-Z]{2}\d{3}", reg_no):
    print("Register Number : Valid")
else:
    print("Register Number : Invalid")
    valid = False

# Email
if re.fullmatch(r"[a-zA-Z0-9._%+-]+@college\.edu", email):
    print("Email : Valid")
else:
    print("Email : Invalid")
    valid = False

# Course Code (Example: CS301)
if re.fullmatch(r"[A-Z]{2}\d{3}", course):
    print("Course Code : Valid")
else:
    print("Course Code : Invalid")
    valid = False

# Semester (1-8)
if re.fullmatch(r"[1-8]", semester):
    print("Semester : Valid")
else:
    print("Semester : Invalid")
    valid = False

# Mobile Number
if re.fullmatch(r"[6-9]\d{9}", mobile):
```